Sahaj Bhadja

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Summary

Early-career software engineer with 2+ years building **high-performance C++/Java systems**, **data-intensive services**, and **production automation** with strong algorithmic problem solving and cross-functional execution.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **performance-focused backend services** for 500+ users, reducing latency from **1.2s to 350ms** by optimizing algorithms, tuning SQL paths, and balancing compute-to-I/O behavior across critical production workflows.
- Implemented **automated validation pipelines** for 5,000+ monthly jobs, improving execution reliability, reducing downstream failures, and strengthening operational visibility for faster incident response and tighter release-quality feedback loops across teams.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **full-stack features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable services and responsive interfaces while improving sprint delivery predictability through clearer task decomposition and implementation sequencing.
- Improved **release velocity** through CI/CD automation and structured defect triage, reducing turnaround by **60%**, lowering manual QA friction, and enabling safer production rollouts across parallel engineering workstreams.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **large-scale experimentation pipelines** across 100,000+ trajectory records, enabling rigorous algorithm evaluation, reproducible benchmarking, and higher-confidence performance comparisons that accelerated iteration speed for research and engineering collaborators.
- Produced **technical reports and test scenarios** translating findings into implementation guidance, improving cross-team alignment, reducing handoff ambiguity, and strengthening execution quality as prototypes advanced toward production-grade standards.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations, resilient data movement, and stricter reliability requirements across production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving developer confidence in frequent releases across coordinated service components.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening real-time operational responsiveness for high-volume event-driven processes.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that slowed validation, troubleshooting, and cross-team handoffs.

Skills

Languages: **C++**, **Java**, Python, SQL, JavaScript/TypeScript, Bash
Core CS: algorithms, data structures, complexity analysis, concurrent programming fundamentals
Backend & Data: REST APIs, distributed services, ETL pipelines, relational databases, query optimization
Tooling: Git/GitHub, Jenkins, Docker, CI/CD, Linux/UNIX workflows, technical documentation
Collaboration: Agile/Scrum execution, cross-functional partnership with researchers, analysts, and product teams

Projects

CPU Scheduling Simulator Code | *C++, React.js, JavaScript*
- Implemented **FCFS, SJF, SRT, RR, and MLQ algorithms** with interactive visualization and metrics tooling, reducing debugging time by **20%** while improving correctness analysis for edge-case scheduling behavior under varied workloads.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving production reliability, reducing deployment variance, and enabling safer iteration across operationally sensitive data-processing workloads.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Developed a **schema-aware query assistant** spanning 10+ datasets with explainable outputs and validation checks, improving analyst speed on ad-hoc tasks while reducing malformed SQL generation and repetitive troubleshooting during exploratory analysis.